

Laboratory work 2

Branched algorithms

Objective : To to acquire skills in the use of conditional operators in the program and the compilation of block diagrams of programs and programs in the C language.

Short theoretical information

2.1. The conditional operator (branching operator) if serves to execute or not execute some operator depending on the truth of certain conditions. The general form of the conditional if statement is as follows:

```
if (expression) operator;  
or  
if (expression) operator;  
else operator;
```

The first form is called the abbreviated, the second is the full form of the conditional operator. If the expression is true (that is, takes any value other than zero), then the statement or block of statements following the if is executed. Otherwise, the statement (or block of statements) following the else is executed. The result of a conditional expression that is included in if can be, for example, an integer, a symbol, a floating-point number, a value of type `_Bool`.

The if statement can be inside another statement if or else. Such an operator is called a nested conditional operator.

Example:

```
if  
{  
    if(b) statement1;  
    if(c) statement2; /* this if statement */  
    else operator3; /* associated with this */  
    /*by the else statement */  
}  
else operator4; /* associated with if(a) */
```

An example of a shortened form of a conditional operator:

```
if (x>y) z = x;
```

An example of the full form of the conditional statement:

```
if (x>y) printf("X is greater than Y"); else  
{  
    z = y; printf("X does not exceed Y");  
}
```

Example 2.1. Complete the task from example 1.4 of the laboratory work №1 with verification of the existence of a triangle.

To solve this problem, it is necessary to check whether the following

conditions are met: the sum of the lengths of the two sides of the triangle must not exceed the length of the third; the lengths entered must be positive. For this, we will use the if operator.

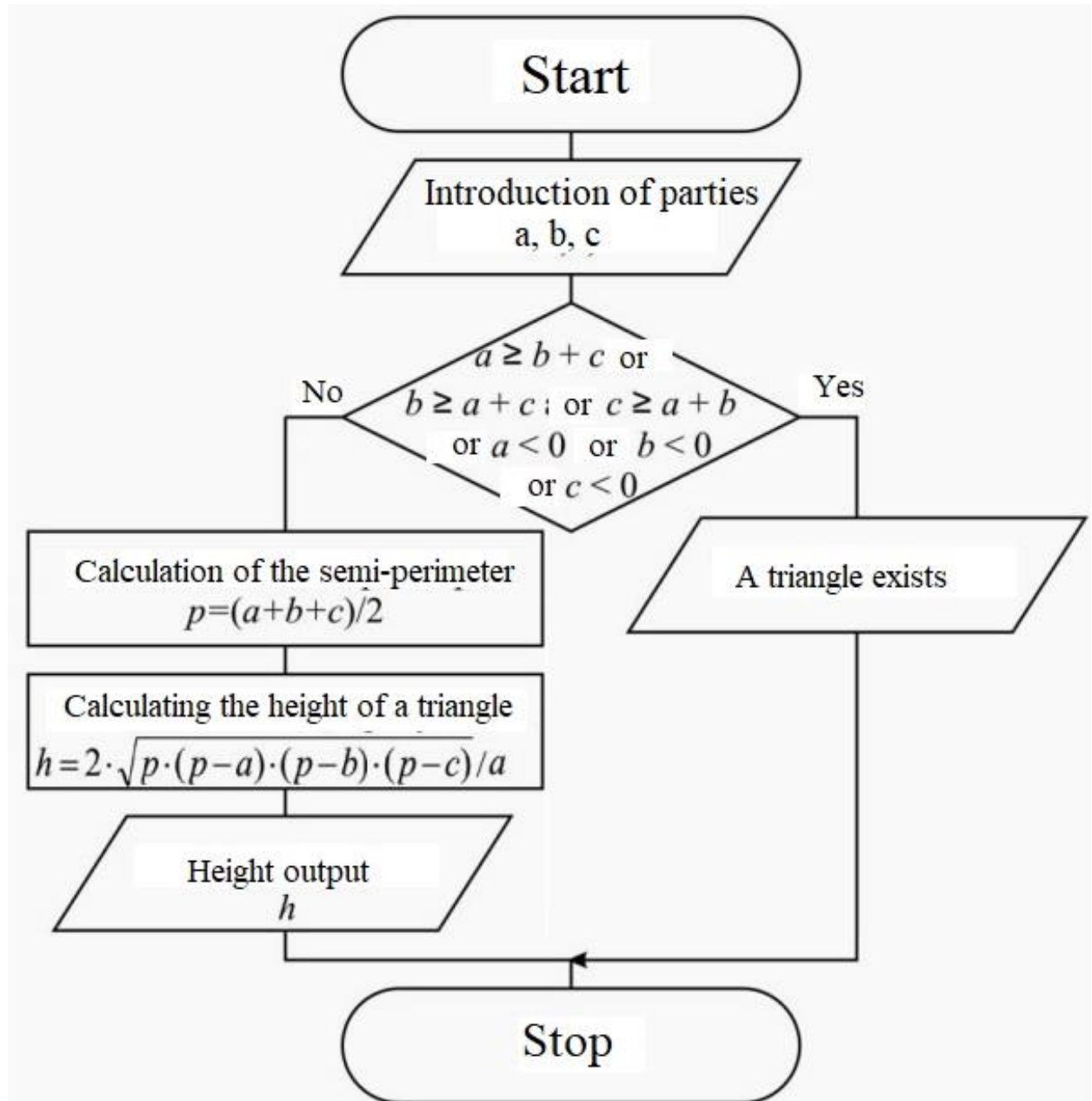


Fig. 2.1 - Block diagram of the algorithm of the program

Program text

```
#define _CRT_SECURE_NO_WARNINGS
#include <stdio.h>
#include <math.h>
#include <locale.h> /* header file used for localization-related tasks*/
int main(void)
{
    int a, b, c;
```

```

float p, ha;
printf("a = ");
scanf("%i", &a);
printf("b = ");
scanf("%i", &b);
printf("c = ");
scanf("%i", &c);
if ((a >= b + c) || (b >= a + c) || (c >= a + b) || (a < 0)
|| (b < 0) || (c < 0))
{
    setlocale(LC_ALL, "ukr"); /*provision of output
    messages in Ukrainian in the console */
    printf("The triangle does not exist!\n");
}
else {
    p = (a + b + c) / 2.0;
    ha = 2*sqrtf(p*(p - a)*(p - b)*(p - c)) / a;
    printf("ha = %8.3f\n", ha);
}
return 0;
}

```

Results of the program:

a = 23

b = 34

c = 45

ha = 33.187

a = -3

b = 4

c = 5

The triangle does not exist!

Example 2.2.

Calculate the largest value from the function values

$y_1 = x^2 + 1$, $y_2 = 7 - x^2$ or $y = x + 1$ for any x .

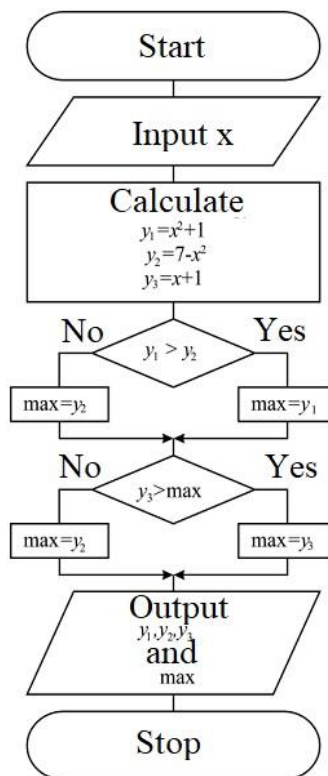


Fig. 2.2 - Block diagram of the algorithm of the program

Program text

```

#define _CRT_SECURE_NO_WARNINGS
#include <stdio.h>
#include <locale.h>
int main(void)
{
    float x, y1, y2, y3, max;
    setlocale(LC_ALL, "ukr");
    printf("Enter x\n");
    scanf("%f", &x);
    y1 = x*x + 1;
    y2 = 7 - x*x;
    y3 = x + 1;
    if (y1 > y2) max = y1; else max = y2;
    if (y3 > max) max = y3;
    printf("y1 = %5.3f y2 = %5.3f y3 = %5.3f\n", y1, y2, y3);
    printf("at x = %6.2f the largest value of the function = %6.2f\n", x, max);
    return 0;
}
  
```

Results of the program:

Enter x

5

y1 = 26,000 y2 = -18,000 y3=6,000

at x = 5.00 the largest value of the function =
26.00

Tasks of laboratory work

Task 1

Write a program that solves the task according to the options.

1. Given two real numbers x and y . Print the number of the quadrant on the coordinate plane to which the point (x, y) belongs.
2. Given five real numbers: a_1, a_2, a_3, a_4, a_5 . It is known that one of them is different from the other four, which are equal to each other.

Find the number of this number.

3. Given numbers a, b, c . Determine whether a triangle with such sides exists.

4. A given triangle with sides a, b, c . Determine what it is:

acute-angled, rectangular, obtuse-angled.

Task 2

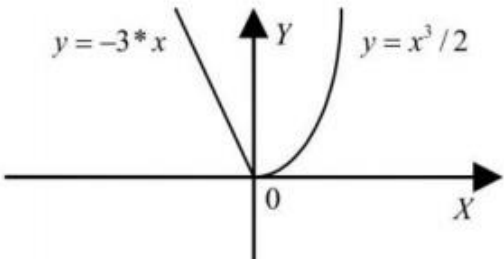
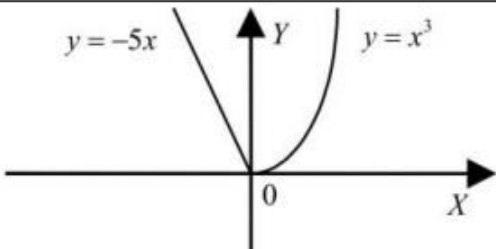
Write a program according to the option.

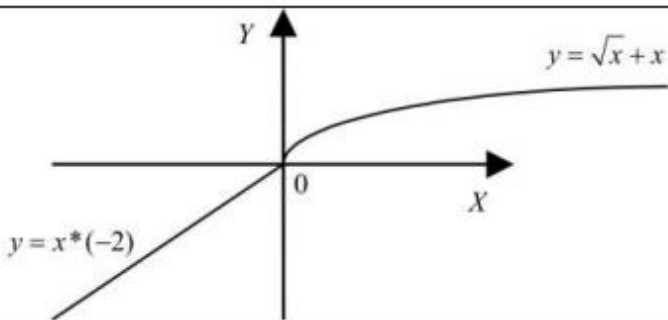
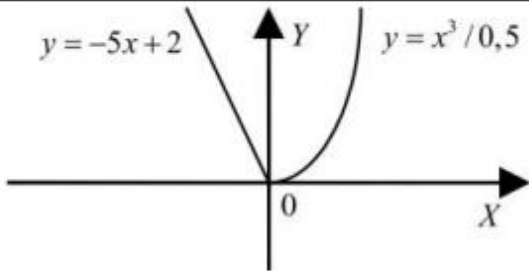
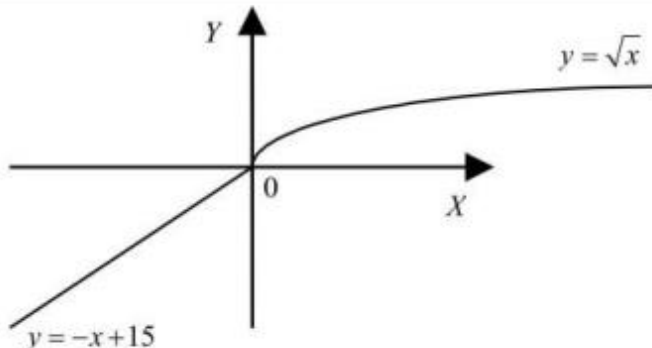
1. The specified number. Print the number of the decade (1–10 – the first; 11–20 – the second; 21–31 – the third).
2. Set temperature. Derive the state in which water is at this temperature.
3. The given angle. Determine its type (sharp, straight, blunt, expanded, more expanded, full).
4. The given age of a person. Write a program that analyzes a person according to his age and assigns him to one of four groups: preschooler, schoolboy, worker, pensioner.
5. Given an integer x . If it is positive, then add 1 to it; if negative, subtract 2 from it; if equal to 0, then replace it with 10.

Task 3.

The function $y = f(x)$ is given graphically, calculate the value of this function (set x independently).

Task options are listed in Table 2.1.

Variant	Function
1	
2	

	
4	
5	

Task 4.

Write a program that solves the task.

Task options:

1. Given a positive three-digit number N. Check whether two of its digits are the same.
2. A positive four-digit number N is specified. Check whether the number 6 is included in the record of this number.
3. Given an integer positive four-digit number N. Determine whether enters the record of the number 0.
4. Given a positive four-digit integer N. Determine the sum the first three digits.
5. Given a four-digit integer N. Determine the sum of the first and last digits.